

Tutorial 8 – Software Design Quality

1. Analyse the requirements given below and provide relevant non-functional requirement. Explain your answer.

- (a) System Y shall process a minimum of 8 transactions per second-**performance**
- (b) The system shall be developed for PC and Macintosh platforms-**portability**
- (c) The system must encrypt all external communications using the RSA algorithm-**security**

2. Given the four 'ilities' reliability, usability, maintainability, reusability, explain how you would rank them in order of importance for each of the following systems:

- (a) A process control system for a chemical plant that produces toxic insecticides

reliability, usability, maintainability, reusability,

- (b) A bank auto teller network used to issue cash and short statements describing accounts

reliability, usability, maintainability, reusability,

- (c) An interactive database used for stock control at each branch of a chain of stores that sells electrical goods.

reliability, usability, reusability, maintainability

- (d) A system for a network of self-service kiosks that can be used to top-up the credit amount in Touch-n-Go cards;

usability, reliability, maintainability, reusability

3. "Every system has real-time performance constraints." Discuss. Or provide a counterexample.

Yes. This may cause by the operation environment such as hardware that running the system. For example , internet bandwidth that used for a web system.

4. Security and usability are often seen to be at odds with each other. Security often imposes procedures and processes that seem like needless overhead to the casual user. But some say that security and usability go (or should go) hand in hand and argue that making the system easy to use securely is the best way to promote security to the user. Discuss.

Usability – provide support to user in using the system's interface. It may assists in improving system's security, for example, applying input mask in a birthdate or mask the password entry.

5. Explain why design conflicts might arise when designing an architecture for which both availability and security requirements are the most important non-functional requirements.

When enforcing availability of system, this may lead to the expose of some core or serious part of the system to users where may lead to possible attack. Some security features may need to close or deny access to some part of system. Therefore, there should be balance on maintaining these two quality dfactors.

6. Many of design decisions for testability are also useful for achieving modifiability. Why do you think that is? Discuss your point of view.

When design is testable, means that there is a measurement for the system's parts to be tested. This will help the developer to know what to modify, the extend of modification. Thus, testability lead to maintainability. If something cannot be tested, how do we know the mofdficaitons?